

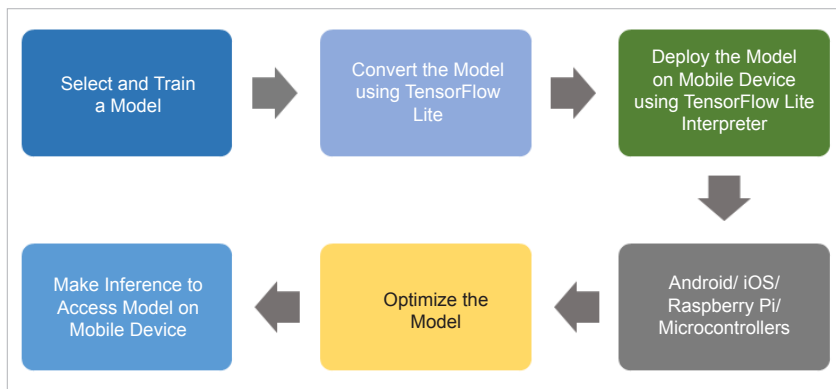
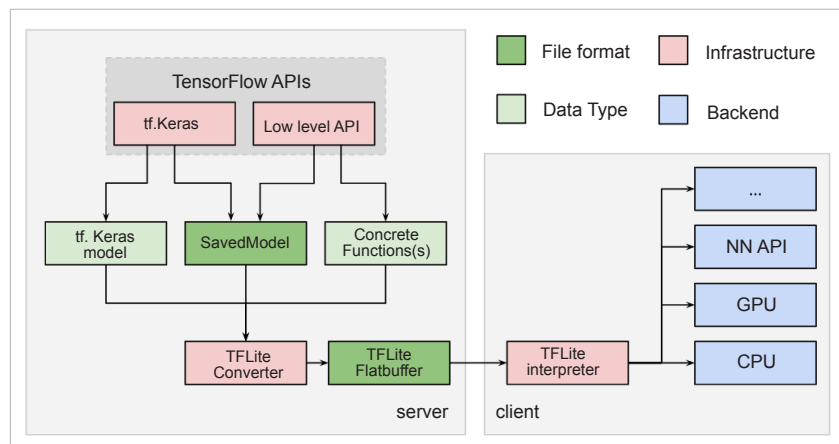


Figure 1: The workflow of TensorFlow Lite

Figure 2: The process of conversion to TensorFlow Lite (Image source: <https://www.tensorflow.org/lite/convert/index>)

The trained model after the training process is required to be saved. The saved model is an architecture that stores different information in a single file. The information is related to weights, biases and training configuration. Sharing and deploying the model becomes easy with the saved model. The following code is used to convert the saved model to TensorFlow Lite:

```
#save the model after compiling
model.save('test_keras_model.h5')
keras_model= tf.keras.models.load_model(' test_keras_model.h5')

#conver the tf.keras model to
TensorFlow Lite model
converter = tf.lite.TFLiteConverter.
```

```
from_keras_model(keras_model)
tflite_model = converter.convert()
```

Model optimisation

An optimised model demands less

space and resources on handhelds like mobile and embedded devices. Therefore, TensorFlow Lite uses two special approaches — quantization and weight pruning — to achieve model optimisation. Quantization makes the model lightweight. It refers to the process of reducing the precision of the numbers that are used to present the different parameters of the TensorFlow model. Weight pruning is the process of trimming those parameters in the model that have less impact on the overall model performance.

Making an inference to access a model on a mobile device

As mentioned above, a TensorFlow Lite model can be deployed on many types of handheld devices. However, for that, the following steps need to be taken to create a mobile interface to access the model:

- With the model, initialise the load interpreter.
- Assign the tensors, and obtain input and output tensors.
- Read and pre-process the image by loading it into a tensor.
- Use and invoke an interpreter that makes the inference on the input tensor.
- Generate the result for the image by mapping the result from the inference. **END** 🐧

References

- [1] <https://www.tensorflow.org/lite>
- [2] <https://towardsdatascience.com/a-basic-introduction-to-tensorflow-lite-59e480c57292>
- [3] <https://towardsdatascience.com/a-beginners-introduction-to-tensorflow-lite-924320deed5>
- [4] Farhoodfar, Avid. (2019). *Machine Learning for Mobile Developers: Tensorflow Lite Framework*

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